

PROBLEM SET 2

COM SCI 188: Intro to Robotics

Spring 2025

Question:	1	2	3	4	5	6	Total
Points:	5	10	5	20	25	35	100

- (5 points) **Reference frame.** Which of the following best describes a reference frame in robotics?
 - A moving or fixed coordinate system for describing positions
 - A physical marker used to measure distances between parts
 - A coordinate system for the position and orientation of robot parts or objects
 - A transformation matrix that stores the robot's movement history
- Homogenous coordinates.** Given a vector $v = [9, 12, 3]$.
 - (5 points) If it is represented a point in Homogenous coordinate. What's its corresponding Cartesian coordinate?
 - (5 points) If it is represented a point in Cartesian coordinate. What's its corresponding Homogenous coordinate?
- (5 points) **Homogeneous transformation.** In a homogeneous 3D transformation matrix, where is the rotation matrix located?
 - Top-left 3×3 block
 - Bottom-right 3×3 block
 - Top-right 3×1 column
 - Bottom-left 3×1 column

4. Rotation matrices.

(a) (4 points) Which axis does the following matrix rotate around?

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

- A) X-axis
- B) Y-axis
- C) Z-axis
- D) Arbitrary axis

(b) (4 points) What is the determinant of a proper rotation matrix in 3D?

- A) 0
- B) 1
- C) -1
- D) Any real number

(c) (4 points) Which of the following is a key property of a valid 3D rotation matrix R ?

- A) $R^T = -R$
- B) $R^T = R^{-1}$
- C) $\det(R) = 0$
- D) $R + R^T = I$

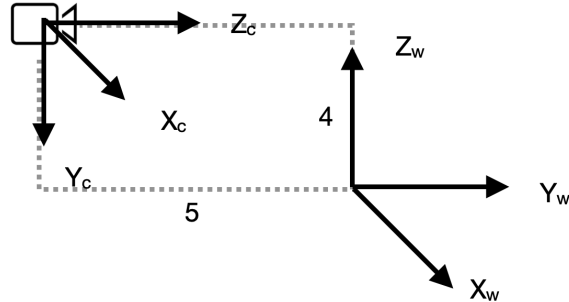
(d) (4 points) What happens when you multiply two rotation matrices $R_1 R_2$?

- A) The result is a translation matrix
- B) The resulting matrix is not a rotation matrix
- C) You get a new rotation representing both rotations applied in sequence
- D) It zeroes out the transformation

(e) (4 points) A rotation matrix must be:

- A) Symmetric and non-singular
- B) Orthogonal with a determinant of ± 1
- C) Orthogonal with a determinant of 1
- D) Skew-symmetric and invertible

5. **Camera Calibration.** Let's defined a world coordinate frame and a camera co-ordinate Frame, as shown in figure below. Camera's position is $[0, -5, 4]$ in world frame, pointing along the Y-axis of the coordinate.



(a) (10 points) What is the camera's extrinsic matrix?

(b) (5 points) If ${}^wP = [-1, 5, 3]$ what is cP

(c) (5 points) If camera's intrinsic matrix is:

$$R = \begin{bmatrix} 100 & 0 & 200 \\ 0 & 100 & 100 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute P's 2D image $[u, v]$ coordinate (aligned with camera).

(d) (5 points) Assume the depth value for pixel $[u, v] = 2.8$. What is the projective signed distance value for location P? (Before truncation and normalization). Is this point P in occluded space or free space?

6. Particle Filter.

- (a) (5 points) What is the main purpose of a particle filter?
- A) To increase the resolution of sensor data
 - B) To smooth out random noise in static images
 - C) To estimate the state of a system using a set of weighted samples
 - D) To detect edges in an image using gradient-based methods
- (b) (5 points) What is the purpose of the “resampling” step in a particle filter?
- A) To increase the number of particles in the system
 - B) To replace low-weight particles with higher-weight ones
 - C) To convert particles into actual measurements
 - D) To reduce noise from the sensor readings
- (c) (5 points) Which of the following is a common use case of particle filters?
- A) Encrypting messages securely over a network
 - B) Detecting malware in software systems
 - C) Estimating the position of a moving robot using noisy sensor data
 - D) Compressing large image files with minimal quality loss
- (d) (5 points) What is “particle degeneracy” in the context of particle filters?
- A) When all particles represent the exact same state
 - B) When too many particles have very low weights, and only a few dominate
 - C) When the system model becomes linear and Gaussian
 - D) When the number of particles increases beyond the computational limit

- (e) (15 points) You are using a particle filter to track the position of a 1D robot. You have 3 particles with the following positions and prior weights:

Particle	Position	Prior Weight
A	4.0	0.3
B	5.0	0.4
C	6.0	0.3

The sensor gives a measurement of $z = 5.2$. The likelihood of each particle's position given the measurement is computed using the Gaussian likelihood function:

$$\text{Likelihood}(x) = \exp\left(-\frac{(z - x)^2}{2\sigma^2}\right)$$

Assume $\sigma = 1.0$.

Question: What are the final normalized weights for particles A, B, and C?